

Darjeeling Himalayan Railway — Tindharia Mechanical Workshop

Technical Maintenance & Overhaul Specification: B-Class 0-4-0ST Locomotives

Section 1.0 — General Dimensions & Weight Distribution

- Gauge: 2 feet 0 inches (610 mm) narrow gauge.
- Working adhesive weight: 14.0 tonnes in full working order with 400 imperial gallons of water and 1.0 tonne coal.
- Maximum axle load: 7.2 tonnes on driving axle.
- Rigid wheelbase: 5 feet 6 inches (1,676 mm), allowing minimum curve negotiation down to 50 feet radius.
- Cylinder bore and stroke: 11 inches diameter by 14 inches stroke.
- Coupled wheel diameter: 2 feet 2 inches (660 mm) on tyre tread.

Section 1.1 — Boiler & Pressure Vessel Specifications

- Maximum operating boiler pressure: 140 psi (9.65 bar).
- Safety valve configuration: Twin 2.0-inch Ramsbottom duplex safety valves set to lift precisely at 142 psi.
- Hydraulic test pressure: 210 psi cold water pressure required every 12 months during annual inspection.
- Fusible plugs: Dual lead-filled bronze alloy plugs in firebox crown plate must be renewed every 6 months or 12,000 miles, whichever comes first.
- Boiler barrel diameter: 3 feet 0.5 inches outside, constructed from 3/8 inch steel plate.
- Firebox heating surface: 40 square feet; tube heating surface: 310 square feet across 84 brass smoke tubes (1.5 inch outer diameter).

Section 2.3 — Brake Gear & Vacuum Apparatus

- Brake system type: Ejector-driven vacuum automatic brake with auxiliary steam brake and hand screw apparatus.
- Vacuum brake operating level: Minimum 18 to 20 inches of mercury (Hg) required before departure from Tindharia, Kurseong, or Darjeeling.
- Brake block to wheel tyre clearance: 1/4 inch (6.35 mm) total clearance per block in released position.
- Brake hanger mounting pin torque: Tighten to 185 Nm (136 ft-lb) with cotter pin positively locked.
- Vacuum brake cylinder: 15-inch rolling ring cylinder mounted under rear bunker.
- Piston stroke travel: Normal operating stroke is 2.5 to 3.0 inches; maximum allowable stroke before manual slack adjustment is 4.0 inches (102 mm).

Section 2.4 — Wheelsets & Tyre Profile Tolerances

- Tyre tread condemnation thickness: Minimum allowable tyre thickness on tread centre line is 1.25 inches (31.8 mm).
- Wheel flange wear limit: Flange thickness must not be less than 13/16 inch (20.6 mm) measured at 9/16 inch below flange crest.
- Wheel gauge back-to-back distance: 1 foot 9.75 inches (+/- 1/16 inch) (552.5 mm).
- Axle box horncheek lateral clearance: 3/32 inch (2.4 mm) maximum allowable side play before liner replacement.

Section 3.2 — Running Gear, Motion & Lubrication

- Valve gear type: Walschaerts external valve gear with inside admission piston valves.
- Eccentric crank pin fastening nut torque: 240 Nm (177 ft-lb).
- Connecting rod big-end brasses running clearance: 0.012 to 0.015 inches (0.30 to 0.38 mm); shim adjustment required if exceeding 0.025 inches.
- Crosshead slipper guide clearance: 0.010 inch nominal clearance; discard shims if clearance exceeds 0.028 inches.
- Steam cylinder lubrication: Mechanical displacement syphon lubricator reservoir filled with steam cylinder oil ISO VG 460.
- Cylinder lubricator feed rate: 4 to 6 drops per minute per cylinder under steam.
- Reversing screw shaft: Pack with Grade 2 Calcium Graphite Grease every 48 operating shed hours.

Section 4.1 — Injectors & Boiler Feed Systems

- Injectors: Two Gresham & Craven No. 5 self-acting combination live-steam injectors installed on cab spectacle plate.
- Minimum injector delivery pressure: 145 psi with boiler working pressure at 135 psi.
- Feed delivery pipe union nut torque: Tighten to 110 Nm (81 ft-lb). Inspect copper washer at every wash-out.
- Boiler check / clack valve lift limit: Maximum allowable lift of the internal bronze clack valve disc is 5/16 inch (7.9 mm). If lift exceeds 3/8 inch due to seat reaming, replace the disc.